

Lemma. Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function s.t. differentiable on (a, b) .

If $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Proof. If f is constant, then there is nothing to prove. Assume f is not constant.

Suppose $f(x) > f(a)$ for some $x \in (a, b)$. Then, by the min-max Theorem, there exists $c \in (a, b)$ such that $f(c) \geq f(x), \forall x \in [a, b]$. Thus, $f'(c) = 0$. $\square$

Mean-Value Theorem.

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function s.t. differentiable on (a, b) .

Then, there exists $c \in [a, b]$ such that $f(b) - f(a) = f'(c)(b - a)$.

Proof. Take $g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) + f(a)$

Then, $g(a) = g(b) = 0$, so $g'(c) = 0$ for some $c \in [a, b]$. $\square$

Generalized Mean-Value Theorem.

Let $f, g: [a, b] \rightarrow \mathbb{R}$ be a continuous function s.t. differentiable on (a, b) .

Then, there exists $c \in [a, b]$ such that

$$g'(c)[f(b) - f(a)] = f'(c)[g(b) - g(a)].$$

Proof. Take $h(x) = g(x)[f(b) - f(a)] - f(x)[g(b) - g(a)]$.

Then, $h(a) = g(a)f(b) - g(a)f(a) - f(a)g(b) + f(a)g(a)$

$$h(b) = g(b)f(b) - g(b)f(a) - f(b)g(b) + f(b)g(a). \quad \square$$

L'Hôpital's Theorem.

Let f, g be differentiable on $B_r(x_0) \subseteq \mathbb{R}$, for some $r > 0$ and x_0 .

If $\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} g(x) = 0$ and $\lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)}$ exists,

then
$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)}.$$

Proof. Given $\epsilon > 0$, there exists $\delta > 0$ such that

$$0 < |t - x_0| < \delta \Rightarrow \left| \frac{f'(t)}{g'(t)} - \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} \right| < \epsilon$$

Fix such t , $\frac{f(t)}{g(t)} = \frac{f(t) - 0}{g(t) - 0} = \frac{f(t) - \lim_{x \rightarrow x_0} f(x)}{g(t) - \lim_{x \rightarrow x_0} g(x)} = \frac{f'(c)}{g'(c)}$ for some $0 < |c - x_0| < \delta$

Now, $\left| \frac{f(t)}{g(t)} - \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} \right| = \left| \frac{f'(c)}{g'(c)} - \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} \right| < \epsilon$ for any $0 < |t - x_0| < \delta$,

so Proved. $\square$

Taylor's Theorem.

Let $f: [a, b] \rightarrow \mathbb{R}$ be given and $n \in \mathbb{N}$ be fixed.

If $f^{(n)}$ exists with continuous on $[a, b]$ and differentiable on (a, b) .

Then, given $x \in [a, b]$, there exists $c_x \in (a, x)$ such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} \cdot (x-a)^k + \frac{f^{(n+1)}(c_x)}{(n+1)!} \cdot (x-a)^{n+1}$$

Proof. Given $x \in [a, b]$, define $F(t) = \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} \cdot (x-t)^k$

and for $g(t) = (t-a)^{n+1}$, $G(t) = \sum_{k=0}^n \frac{g^{(k)}(t)}{k!} \cdot (x-t)^k$.

$$\begin{aligned} \text{Then, } F'(t) &= \frac{d}{dt} \left(\sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (x-t)^k \right) \\ &= \sum_{k=0}^n \frac{d}{dt} \left(\frac{f^{(k)}(t)}{k!} \cdot (x-t)^k \right) = \left(\frac{f^{(k+1)}(t)}{k!} \cdot (x-t)^k - \frac{f^{(k)}(t)}{(k-1)!} \cdot (x-t)^{k-1} \right) \\ &= \frac{f^{(n+1)}(t)}{n!} \cdot (x-t)^n \quad (\text{Telescoping}) \end{aligned}$$

$$\text{And } G'(t) = (n+1) \cdot (x-t)^n \quad (g'(t) = (n+1)(t-a)^n, \quad g^{(2)}(t) = (n+1) \cdot n \cdot (t-a)^{n-1} \dots, \quad g^{(n+1)}(t) = (n+1)!))$$

Now, by the generalized Mean-Value Theorem,

$$\underbrace{F'(c_x)}_{//} \cdot \underbrace{[G(x) - G(a)]}_{\searrow} = \underbrace{G'(c_x)}_{\searrow} \cdot \underbrace{[F(x) - F(a)]}_{\searrow}$$

$$\frac{f^{(n+1)}(c_x)}{n!} \cdot (x-c_x)^n \cdot [(x-a)^{n+1} - 0] = (n+1) \cdot (x-c_x)^n \cdot \left[f(x) - \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} \cdot (x-a)^k \right]$$

So, proved.

Some Application of Taylor.

Prop. Let $f: (a, \infty) \rightarrow \mathbb{R}$ be a twice-differentiable function and

$$M_2 \stackrel{\text{def}}{=} \sup_{x > a} |f''(x)| \text{ exists.}$$

Then, $M_1^2 \leq 4 \cdot M_0 \cdot M_2$. Moreover, if $f(x) \rightarrow 0$ as $x \rightarrow \infty$, then $f'(x) \rightarrow 0$ as $x \rightarrow \infty$.

Proof. Let $x \in (a, \infty)$ be fixed. Given $h > 0$, the Taylor's Theorem gives

$$f(x+2h) = f(x) + f'(x) \cdot 2h + \frac{f''(c_x)}{2!} \cdot (2h)^2 \text{ for some } c_x \in (x, x+2h).$$

$$\text{So, } f'(x) = \frac{1}{2h} \cdot (f(x+2h) - f(x)) - h \cdot f''(c_x), \text{ so}$$

$$|f'(x)| = \frac{1}{2h} \cdot [|f(x+2h)| + |f(x)|] + h \cdot |f''(c_x)| \leq \frac{1}{2h} \cdot 2 \cdot M_0 + h \cdot M_2 = \frac{M_0}{h} + h M_2.$$

$$\text{Now, take } h = \sqrt{M_0/M_2}, \text{ then } \frac{M_0}{h} + h M_2 = \sqrt{M_0} \cdot \sqrt{M_2} + \sqrt{M_0} \sqrt{M_2} = 2 \cdot \sqrt{M_0 M_2}.$$

(If $M_2 = 0$, the f' is constant, so f is $f'(0) \cdot x + b$, $f'(0)$ must be zero.)

The second assertion is proved by:

$$\sup_{x > a} |f'(x)| \leq \sup_{x > a} |f''(x)| \cdot \sup_{x > a} |f(x)| \rightarrow 0 \text{ as } a \rightarrow \infty.$$